

SOLIDWORKS API (.NET VBA C#) — Reference (205/337): IModeler (part 3 of 4)

SOLIDWORKS API Documentation (converted from CHM)

IModeler (part 3 of 4) (Type)

SOLIDWORKS API Help

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ICreateBrepBody3 Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : ICreateBrepBody3 Method (IModeler)

Creates a temporary body from BREP (boundary representation) data.

.NET Syntax

Visual Basic (Declaration)

```
Function ICreateBrepBody3( _  
    ByVal Type As System.Integer, _  
    ByVal NTopologies As System.Integer, _  
    ByRef Topologies As System.Integer, _  
    ByRef EdgeTolArray As System.Double, _  
    ByRef VertexTolArray As System.Double, _  
    ByRef PointArray As System.Double, _  
    ByRef CurveArra1 As Curve, _  
    ByRef CurveSurfaceArray1 As Surface, _  
    ByRef CurveArray2 As Curve, _  
    ByRef CurveSurfaceArray2 As Surface, _  
    ByRef SurfaceArray As Surface, _  
    ByVal NRelations As System.Integer, _  
    ByRef Parents As System.Integer, _  
    ByRef Children As System.Integer, _  
    ByRef Senses As System.Integer, _  
    ByVal Option As System.Integer _  
) As Body2
```

Visual Basic (Usage)



```
Dim instance As IModeler  
Dim Type As System.Integer  
Dim NTopologies As System.Integer  
Dim Topologies As System.Integer  
Dim EdgeTolArray As System.Double  
Dim VertexTolArray As System.Double  
Dim PointArray As System.Double  
Dim CurveArra1 As Curve  
Dim CurveSurfaceArray1 As Surface  
Dim CurveArray2 As Curve  
Dim CurveSurfaceArray2 As Surface  
Dim SurfaceArray As Surface  
Dim NRelations As System.Integer  
Dim Parents As System.Integer  
Dim Children As System.Integer  
Dim Senses As System.Integer  
Dim Option As System.Integer  
Dim value As Body2
```

```
value = instance.ICreateBrepBody3(Type, NTopologies, Topologies, EdgeTolArray, VertexTolArray,  
PointArray, CurveArra1, CurveSurfaceArray1, CurveArray2, CurveSurfaceArray2, SurfaceArray,  
NRelations, Parents, Children, Senses, Option)
```

C#

```
Body2 ICreateBrepBody3(  
    System.int Type,  
    System.int NTopologies,  
    ref System.int Topologies,  
    ref System.double EdgeTolArray,  
    ref System.double VertexTolArray,  
    ref System.double PointArray,  
    ref Curve CurveArra1,
```

```

    ref Surface CurveSurfaceArray1,
    ref Curve CurveArray2,
    ref Surface CurveSurfaceArray2,
    ref Surface SurfaceArray,
    System.int NRelations,
    ref System.int Parents,
    ref System.int Children,
    ref System.int Senses,
    System.int Option
)

```

C++/CLI

```

Body2^ ICreateBrepBody3(
& System.int Type,
& System.int NTopologies,
& System.int% Topologies,
& System.double% EdgeTolArray,
& System.double% VertexTolArray,
& System.double% PointArray,
& Curve^% CurveArra1,
& Surface^% CurveSurfaceArray1,
& Curve^% CurveArray2,
& Surface^% CurveSurfaceArray2,
& Surface^% SurfaceArray,
& System.int NRelations,
& System.int% Parents,
& System.int% Children,
& System.int% Senses,
& System.int Option
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Type

Type of body to create as defined in [swTopology_e](#)

NTopologies

Number of topological entities in the topologies argument

Topologies

Array of topologies (see [swTopoEntity_e](#)), one for each topological entity

EdgeTolArray

Array of tolerances for edges

VertexTolArray

Array of tolerances for vertices

PointArray

Array of coordinates of vertices (geometry for vertices)

CurveArra1

Array of [curves](#) (geometry for edges; 3D curve) and coedges (geometry for coedges; 2D curve)

CurveSurfaceArray1

Array of [surfaces](#) that lie on CurveArra1 2D curve

CurveArray2

Array of [curves](#) (geometry for edges; 3D curve) and array of coedges (geometry for coedges; 2D curve)

CurveSurfaceArray2

Array of [surfaces](#) that on lie on CurveArray2 2D curve

SurfaceArray

Array of [surfaces](#) (geometry for faces)

NRelations

Number of 1-to-1 relationships between topological entities

Parents

Array of parents, one in each relationship

Children

Array of children, one in each relationship

Senses

Array of senses in which child is used by parent in the relationship

Option

• 0 = Default

- 1 = Repair and simplify body
- 2 = Simplify body

Return Value

[Body](#)

Visual Basic for Applications (VBA) Syntax

See [Modeler::ICreateBrepBody3](#).

Remarks

The CurveArray1 and CurveArray2 must be paired with CurveSurfaceArray1 and CurveSurfaceArray2, respectively. Order is not important. The 2D curve has to be created in the direction of the loop.

If non-negative values are packed into the EdgeToleranceArray and VertexToleranceArray arrays, then tolerances are applied to the corresponding edges or vertices. These arrays should be the same size as CurveArray1 and PointArray, respectively. Otherwise, a default value of 1.0e-8 (modeler precision) is used.

NOTE: [IModeler::SetInitKnitGapWidth](#) does not affect this method.

Useful methods for creating geometry for the topological entities are:

- [IBody2::CreatePlanarSurface](#) or [IBody2::ICreatePlanarSurface](#)
- [IBody2::AddProfileArc](#) or [IBody2::IAddProfileArc](#)
- [IBody2::AddProfileLine](#) or [IBody2::IAddProfileLine](#)
- [IBody2::CreateRevolutionSurface](#) or [IBody2::ICreateRevolutionSurface](#)

For example, to create a cone, find topological relationships and form relevant arrays. A complete solid cone consists of 8 topological entities:

- 1 shell
- 2 faces
- 3 loops
- 1 edge
- 1 vertex

There are 8 relations:

- the shell is the parent of 2 faces - 2
- the planar end face has 1 loop - 1
- the conical face has 2 loops - 2
- two loops are each the parent of 1 edge - 2
- one loop is the parent of 1 vertex 1

The topologies array:

Index	Value
0	swTopoShell
1	swTopoFace
2	swTopoFace
3	swTopoLoop
4	swTopoEdge
5	swTopoLoop
6	swTopoLoop
7	swTopoVertex

The set of arrays:

index	parents	children	senses	relation
0	0	1	0	shell to face
1	0	2	0	shell to face
2	1	3	0	face to loop
3	3	4	-1	loop to edge
4	2	5	0	face to loop
5	2	6	0	face to loop
6	5	4	1	loop to edge
7	6	7	0	loop to vertex

Values in the parents and children arrays correspond to the indices of the topology array.

Each face or edge created by [IModeler::CreateBrepBody3](#) or `IModeler::ICreateBrepBody3` has an associated integer ID that is the same as the index to the input topologies. Use [IFace2::GetFaceId](#) or [IEdge::GetId](#) to get the associated integer ID.

Every shell should be a closed shell. Sheet bodies should have additional back faces to form a closed shell. When creating a sheet body, these extra back faces are retained in the result and should be removed using [IModeler::DeleteFacesFromSheetBody](#) or [IModeler::IDeleteFacesFromSheetBody](#).

See Also

[IModeler Interface](#)
[IModeler Members](#)
[IModeler::CreateBodiesFromSheets2 Method](#)
[IModeler::CreateBodyFromBox Method](#)
[IModeler::CreateBodyFromCone Method](#)
[IModeler::CreateBodyFromCyl Method](#)
[IModeler::CreateBodyFromFaces2 Method](#)
[IModeler::CreateBrepBody3 Method](#)
[IModeler::CreateExtrudedBody Method](#)
[IModeler::CreateSweptBody Method](#)
[IModeler::CreateWireBody Method](#)
[IModeler::ICreateBodiesFromSheets2 Method](#)
[IModeler::ICreateBodyFromBox2 Method](#)
[IModeler::ICreateBodyFromCone2 Method](#)
[IModeler::ICreateBodyFromCyl2 Method](#)
[IModeler::ICreateBodyFromFaces3 Method](#)
[IModeler::ICreateWireBody Method](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ICreateBsplineCurve Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : ICreateBsplineCurve Method (IModeler)

Creates a b-spline curve.

.NET Syntax

Visual Basic (Declaration)

```
Function ICreateBsplineCurve( _  
    ByRef Properties As System.Integer, _  
    ByRef KnotArray As System.Double, _  
    ByRef ControlPointCoordArray As System.Double _  
) As Curve
```

Visual Basic (Usage)



```
Dim instance As IModeler  
Dim Properties As System.Integer  
Dim KnotArray As System.Double  
Dim ControlPointCoordArray As System.Double  
Dim value As Curve  
  
value = instance.ICreateBsplineCurve(Properties, KnotArray, ControlPointCoordArray)
```

C#

```
Curve ICreateBsplineCurve(  
    ref System.int Properties,  
    ref System.double KnotArray,  
    ref System.double ControlPointCoordArray  
)
```

C++/CLI

```
Curve^ ICreateBsplineCurve(  
& System.int% Properties,  
& System.double% KnotArray,
```

```
& System.double% ControlPointCoordArray
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Properties

Array containing 4 integers packed into 2 double elements (see **Remarks**)

KnotArray

Array of numKnots doubles (see **Remarks**)

ControlPointCoordArray

Array of NumCtrlPtCoord doubles (see **Remarks**)

Return Value

B-spline [curve](#)

Visual Basic for Applications (VBA) Syntax

See [Modeler::ICreateBsplineCurve](#).

Remarks

The Properties argument contains the following values:

- DimensionControlPoints
- Order
- NumCtrlPoints
- Periodicity

Length of this array...	Given by...
Knots	numKnots = NumCtrlPoints + Order
CtrlPtCoords	NumCtrlPtCoord = NumCtrlPoints + DimensionControlPoints

See Also

[IModeler Interface](#)

[IModeler Members](#)

[IModeler::CreateBsplineCurve Method](#)

[IModeler::CreateBsplineSurface Method](#)

[IModeler::ICreateBsplineSurface Method](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

ICreateBsplineSurface Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : ICreateBsplineSurface Method (IModeler)

Creates a b-spline surface.

.NET Syntax

Visual Basic (Declaration)

```
Function ICreateBsplineSurface( _
    ByRef Properties As System.Integer, _
    ByRef UKnotArray As System.Double, _
    ByRef VKnotArray As System.Double, _
    ByRef ControlPointCoordArray As System.Double _
) As Surface
```

Visual Basic (Usage)

```
Dim instance As IModeler
Dim Properties As System.Integer
Dim UKnotArray As System.Double
Dim VKnotArray As System.Double
Dim ControlPointCoordArray As System.Double
Dim value As Surface
```



```
value = instance.ICreateBsplineSurface(Properties, UKnotArray, VKnotArray, ControlPointCoordArray)
```

C#

```
Surface ICreateBsplineSurface(  
    ref System.int Properties,  
    ref System.double UKnotArray,  
    ref System.double VKnotArray,  
    ref System.double ControlPointCoordArray  
)
```

C++/CLI

```
Surface^ ICreateBsplineSurface(  
& System.int% Properties,  
& System.double% UKnotArray,  
& System.double% VKnotArray,  
& System.double% ControlPointCoordArray  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Properties
Array of 8 integers (see **Remarks**)
UKnotArray
Array of numUKnots (see Remarks)
VKnotArray
Array of numVKnots (see Remarks)
ControlPointCoordArray
Array of numCtrlPtCoord (see Remarks)

Return Value

[Surface](#)

 Visual Basic for Applications (VBA) Syntax

See [Modeler::ICreateBsplineSurface](#).

 Remarks

Number of elements in this array...	Given by...
UKnots	numUKnots = numU_CtrlPoints + Uorder
VKnots	numVKnots = numV_CtrlPoints + Vorder
CtrlPtCoords	numCtrlPtCoord =(NumV_CtrlPoints * NumU_CtrlPoints * DimensionControlPoints)

NOTE: The order of the UV control points are reversed if you used [ISurface::GetBSurfParams3](#) or [ISurface::IGetBSurfParams](#) to get the data. You do not need to reverse the UV control points; instead, you can switch the UV knots and related parameters.

For periodics, the first knot must have multiplicity = 1 and all excess multiplicity must be imposed on the last knot. Therefore, a valid periodic knv vector would be { 0, 1, 2, 3, 3, 3 } for a cubic curve. The control point on the seam of the closed curve cannot be replicated at both ends; it should occur only at the beginning.

To convert from a clamped periodic curve (numKnots = numCtrlPts + Order, ctrlPts replicated, knot multiplicity = order at each end) to a SOLIDWORKS periodic curve, remove all but one of the knots at the beginning of the vector and remove one at the end. Also remove the last control point to avoid the point replication.

 See Also

[IModeler Interface](#)
[IModeler Members](#)
[IModeler::CreateBsplineSurface Method](#)
[IBody2::CreateBsplineSurface Method](#)
[IBody2::ICreateBsplineSurface Method](#)
[IModeler::CreateBsplineCurve Method](#)
[IModeler::ICreateBsplineCurve Method](#)
[IModeler::CreateConicalSurface2 Method](#)
[IModeler::CreateCoonsBSurface Method](#)
[IModeler::CreateCylindricalSurface2 Method](#)
[IModeler::CreateExtrusionSurface Method](#)
[IModeler::CreateLoftSurface Method](#)
[IModeler::CreateOffsetSurface Method](#)

[IModeler::CreatePlanarSurface2 Method](#)
[IModeler::CreateRuledSurface Method](#)
[IModeler::CreateSweptSurface Method](#)
[IModeler::CreateToroidalSurface Method](#)
[IModeler::ICreateConicalSurface2 Method](#)
[IModeler::ICreateCylindricalSurface2 Method](#)
[IModeler::ICreateExtrusionSurface Method](#)
[IModeler::ICreateLoftSurface Method](#)
[IModeler::ICreateOffsetSurface Method](#)
[IModeler::ICreatePlanarSurface2 Method](#)
[IModeler::ICreateRuledSurface Method](#)
[IModeler::ICreateSweptSurface Method](#)
[IModeler::ICreateToroidalSurface Method](#)

SOLIDWORKS API Help

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ICreateConicalSurface Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : ICreateConicalSurface Method (IModeler)

Obsolete. Superseded by [IModeler::ICreateConicalSurface2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function ICreateConicalSurface( _  
    ByRef Center As System.Double, _  
    ByRef Direction As System.Double, _  
    ByVal Radius As System.Double, _  
    ByVal SemiAngle As System.Double _  
) As Surface
```

Visual Basic (Usage)



```
Dim instance As IModeler  
Dim Center As System.Double  
Dim Direction As System.Double  
Dim Radius As System.Double  
Dim SemiAngle As System.Double  
Dim value As Surface
```

```
value = instance.ICreateConicalSurface(Center, Direction, Radius, SemiAngle)
```

C#

```
Surface ICreateConicalSurface(  
    ref System.double Center,  
    ref System.double Direction,  
    System.double Radius,  
    System.double SemiAngle  
)
```

C++/CLI

```
Surface^ ICreateConicalSurface(  
& System.double% Center,  
& System.double% Direction,  
& System.double Radius,  
& System.double SemiAngle  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Center

Direction

Radius

SemiAngle

Visual Basic for Applications (VBA) Syntax

See [Modeler::ICreateConicalSurface](#).

See Also

[IModeler Interface](#)
[IModeler Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

ICreateConicalSurface2 Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : ICreateConicalSurface2 Method (IModeler)

Creates an untrimmed conical surface.

.NET Syntax

Visual Basic (Declaration)

```
Function ICreateConicalSurface2( _  
    ByRef Center As System.Double, _  
    ByRef Direction As System.Double, _  
    ByRef RefDirection As System.Double, _  
    ByVal Radius As System.Double, _  
    ByVal SemiAngle As System.Double _  
    ) As Surface
```

Visual Basic (Usage)



```
Dim instance As IModeler  
Dim Center As System.Double  
Dim Direction As System.Double  
Dim RefDirection As System.Double  
Dim Radius As System.Double  
Dim SemiAngle As System.Double  
Dim value As Surface  
  
value = instance.ICreateConicalSurface2(Center, Direction, RefDirection, Radius, SemiAngle)
```

C#

```
Surface ICreateConicalSurface2(  
    ref System.double Center,  
    ref System.double Direction,  
    ref System.double RefDirection,  
    System.double Radius,  
    System.double SemiAngle  
)
```

C++/CLI

```
Surface^ ICreateConicalSurface2(  
& System.double% Center,  
& System.double% Direction,  
& System.double% RefDirection,  
& System.double Radius,  
& System.double SemiAngle  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Center
Array containing 3 doubles (see **Remarks**)
Direction
Array of 3 doubles (see **Remarks**)
RefDirection
Array containing 3 doubles (see **Remarks**)
Radius
See **Remarks**
SemiAngle
See **Remarks**

Return Value

 Visual Basic for Applications (VBA) Syntax

See [Modeler::ICreateConicalSurface2](#).

 Remarks

Input arguments:

Argument	Description
Center	XYZ location which represents the center of the bottom
Direction	XYZ direction of the axis of the conical surface
Radius	Radius at the center
SemiAngle	Half angle of the cone in radians

You can trim the resulting surface to generate a sheet body (for example, using [ISurface::CreateTrimmedSheet](#)).

 See Also

[IModeler Interface](#)
[IModeler Members](#)
[IModeler::CreateBsplinesSurface Method](#)
[IModeler::CreateCoonsBSurface Method](#)
[IModeler::CreateCylindricalSurface2 Method](#)
[IModeler::CreateExtrusionSurface Method](#)
[IModeler::CreateLoftSurface Method](#)
[IModeler::CreateOffsetSurface Method](#)
[IModeler::CreatePlanarSurface2 Method](#)
[IModeler::CreateRuledSurface Method](#)
[IModeler::CreateSphericalSurface2 Method](#)
[IModeler::CreateSweptSurface Method](#)
[IModeler::CreateToroidalSurface Method](#)
[IModeler::ICreateBsplinesSurface Method](#)
[IModeler::ICreateCylindricalSurface2 Method](#)
[IModeler::ICreateExtrusionSurface Method](#)
[IModeler::ICreateLoftSurface Method](#)
[IModeler::ICreateOffsetSurface Method](#)
[IModeler::ICreatePlanarSurface2 Method](#)
[IModeler::ICreateRuledSurface Method](#)
[IModeler::ICreateSweptSurface Method](#)
[IModeler::ICreateToroidalSurface Method](#)

 Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ICreateCylindricalSurface Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : ICreateCylindricalSurface Method (IModeler)

Obsolete. Superseded by [IModeler::ICreateCylindricalSurface2](#).

 .NET Syntax

Visual Basic (Declaration)

```
Function ICreateCylindricalSurface( _  
    ByRef Center As System.Double, _  
    ByRef Direction As System.Double, _  
    ByVal Radius As System.Double _  
) As Surface
```

Visual Basic (Usage)



```
Dim instance As IModeler  
Dim Center As System.Double  
Dim Direction As System.Double  
Dim Radius As System.Double
```

```
Dim value As Surface
```

```
value = instance.ICreateCylindricalSurface(Center, Direction, Radius)
```

C#

```
Surface ICreateCylindricalSurface(  
    ref System.double Center,  
    ref System.double Direction,  
    System.double Radius  
)
```

C++/CLI

```
Surface^ ICreateCylindricalSurface(  
& System.double% Center,  
& System.double% Direction,  
& System.double Radius  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Center

Direction

Radius

Visual Basic for Applications (VBA) Syntax

See [Modeler::ICreateCylindricalSurface](#).

See Also

[IModeler Interface](#)
[IModeler Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

ICreateCylindricalSurface2 Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : ICreateCylindricalSurface2 Method (IModeler)

Creates an untrimmed cylindrical surface.

.NET Syntax

Visual Basic (Declaration)

```
Function ICreateCylindricalSurface2( _  
    ByRef Center As System.Double, _  
    ByRef Direction As System.Double, _  
    ByRef RefDirection As System.Double, _  
    ByVal Radius As System.Double _  
) As Surface
```

Visual Basic (Usage)

```
Dim instance As IModeler  
Dim Center As System.Double  
Dim Direction As System.Double  
Dim RefDirection As System.Double  
Dim Radius As System.Double  
Dim value As Surface  
  
value = instance.ICreateCylindricalSurface2(Center, Direction, RefDirection, Radius)
```

C#



```
Surface ICreateCylindricalSurface2(  
    ref System.double Center,  
    ref System.double Direction,  
    ref System.double RefDirection,  
    System.double Radius  
)
```

C++/CLI

```
Surface^ ICreateCylindricalSurface2(  
& System.double% Center,  
& System.double% Direction,  
& System.double% RefDirection,  
& System.double Radius  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Center

Array containing 3 doubles for XYZ location of the center of the bottom

Direction

Array containing 3 doubles for XYZ direction of the axis of the cylindrical surface

RefDirection

Array containing 3 doubles for XYZ direction of the axis of the cylindrical surface

Radius

Radius at the center

Return Value

[Surface](#)

Visual Basic for Applications (VBA) Syntax

See [Modeler::ICreateCylindricalSurface2](#).

Remarks

You can trim the resulting surface to generate a sheet body, for example, using [ISurface::CreateTrimmedSheet](#).

See Also

[IModeler Interface](#)

[IModeler Members](#)

[IModeler::CreateBsplinesSurface Method](#)

[IModeler::CreateConicalSurface2 Method](#)

[IModeler::CreateCoonsBSurface Method](#)

[IModeler::CreateCylindricalSurface2 Method](#)

[IModeler::CreateExtrusionSurface Method](#)

[IModeler::CreateLoftSurface Method](#)

[IModeler::CreateOffsetSurface Method](#)

[IModeler::CreatePlanarSurface2 Method](#)

[IModeler::CreateRuledSurface Method](#)

[IModeler::CreateSphericalSurface2 Method](#)

[IModeler::CreateSweptSurface Method](#)

[IModeler::CreateToroidalSurface Method](#)

[IModeler::ICreateBsplinesSurface Method](#)

[IModeler::ICreateConicalSurface2 Method](#)

[IModeler::ICreateExtrusionSurface Method](#)

[IModeler::ICreateLoftSurface Method](#)

[IModeler::ICreateOffsetSurface Method](#)

[IModeler::ICreatePlanarSurface2 Method](#)

[IModeler::ICreateRuledSurface Method](#)

[IModeler::ICreateSphericalSurface2 Method](#)

[IModeler::ICreateSweptSurface Method](#)

[IModeler::ICreateToroidalSurface Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ICreateEllipse Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : ICreateEllipse Method (IModeler)

Creates a temporary elliptical curve.

Visual Basic (Declaration)

```
Function ICreateEllipse( _
    ByRef Center As System.Double, _
    ByVal MajorRadius As System.Double, _
    ByVal MinorRadius As System.Double, _
    ByRef MajorAxis As System.Double, _
    ByRef MinorAxis As System.Double _
) As Curve
```

Visual Basic (Usage)



```
Dim instance As IModeler
Dim Center As System.Double
Dim MajorRadius As System.Double
Dim MinorRadius As System.Double
Dim MajorAxis As System.Double
Dim MinorAxis As System.Double
Dim value As Curve

value = instance.ICreateEllipse(Center, MajorRadius, MinorRadius, MajorAxis, MinorAxis)
```

C#

```
Curve ICreateEllipse(
    ref System.double Center,
    System.double MajorRadius,
    System.double MinorRadius,
    ref System.double MajorAxis,
    ref System.double MinorAxis
)
```

C++/CLI

```
Curve^ ICreateEllipse(
    & System.double% Center,
    & System.double MajorRadius,
    & System.double MinorRadius,
    & System.double% MajorAxis,
    & System.double% MinorAxis
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Center

- in-process, unmanaged C++: Pointer to an array of 3 doubles describing the location of the center of the ellipse
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

MajorRadius

Major radius of ellipse

MinorRadius

Minor radius of ellipse

MajorAxis

- in-process, unmanaged C++: Pointer to an array of 3 doubles describing the major axis of the ellipse
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

MinorAxis

- in-process, unmanaged C++: Pointer to an array of 3 doubles describing the minor axis of the ellipse
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Return Value

[Curve](#)

See Also

[IModeler Interface](#)
[IModeler Members](#)
[IModeler::CreateEllipse Method](#)
[IModeler::CreateArc Method](#)
[IModeler::ICreateArc Method](#)
[IModeler::CreateLine Method](#)
[IModeler::ICreateLine Method](#)
[IModeler::CreateExtrudedBody Method](#)

Availability

SOLIDWORKS 2012 SP01, Revision Number 20.1

SOLIDWORKS API Help

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ICreateExtrusionSurface Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : ICreateExtrusionSurface Method (IModeler)

Creates an extruded surface.

.NET Syntax

Visual Basic (Declaration)

```
Function ICreateExtrusionSurface( _  
    ByVal ProfileCurve As Curve, _  
    ByRef AxisDirection As System.Double _  
) As Surface
```

Visual Basic (Usage)



```
Dim instance As IModeler  
Dim ProfileCurve As Curve  
Dim AxisDirection As System.Double  
Dim value As Surface  
  
value = instance.ICreateExtrusionSurface(ProfileCurve, AxisDirection)
```

C#

```
Surface ICreateExtrusionSurface(  
    Curve ProfileCurve,  
    ref System.double AxisDirection  
)
```

C++/CLI

```
Surface^ ICreateExtrusionSurface(  
& Curve^ ProfileCurve,  
& System.double% AxisDirection  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ProfileCurve
Profile [curve](#)
AxisDirection
Array of 3 doubles (x,y,z)

Visual Basic for Applications (VBA) Syntax

See [Modeler::ICreateExtrusionSurface](#).

Remarks

You can use this method with:

- A set of related functions that construct a body from trimmed surfaces. The profile curve is extruded along the direction vector of axis direction, the new surface being the envelope of the curve. The profile curve must be of type line, circle, or b-spline curve.
- Trimming curve creation routines (for example, [ISurface::AddTrimmingLoop2](#) or [ISurface::IAddTrimmingLoop2](#)) to construct a trimmed tabulated cylinder.

See Also

[IModeler Interface](#)
[IModeler Members](#)
[IModeler::CreateExtrusionSurface Method](#)
[IBody2::CreateExtrusionSurface Method](#)
[IBody2::ICreateExtrusionSurface Method](#)
[IModeler::CreateBsplinesSurface Method](#)
[IModeler::CreateConicalSurface2 Method](#)
[IModeler::CreateCoonsBSurface Method](#)
[IModeler::CreateCylindricalSurface2 Method](#)
[IModeler::CreateLoftSurface Method](#)
[IModeler::CreateOffsetSurface Method](#)
[IModeler::CreatePlanarSurface2 Method](#)
[IModeler::CreateRuledSurface Method](#)
[IModeler::CreateSweptSurface Method](#)
[IModeler::CreateToroidalSurface Method](#)
[IModeler::ICreateBsplinesSurface Method](#)
[IModeler::ICreateConicalSurface2 Method](#)
[IModeler::ICreateCylindricalSurface2 Method](#)
[IModeler::ICreateLoftSurface Method](#)
[IModeler::ICreateOffsetSurface Method](#)
[IModeler::ICreatePlanarSurface2 Method](#)
[IModeler::ICreateRuledSurface Method](#)
[IModeler::ICreateSphericalSurface2 Method](#)
[IModeler::ICreateSweptSurface Method](#)
[IModeler::ICreateToroidalSurface Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

ICreateLine Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : ICreateLine Method (IModeler)

Creates a line.

.NET Syntax

Visual Basic (Declaration)

```
Function ICreateLine( _
    ByRef RootPoint As System.Double, _
    ByRef Direction As System.Double _
) As Curve
```

Visual Basic (Usage)

```
Dim instance As IModeler
Dim RootPoint As System.Double
Dim Direction As System.Double
Dim value As Curve

value = instance.ICreateLine(RootPoint, Direction)
```

C#

```
Curve ICreateLine(
    ref System.double RootPoint,
    ref System.double Direction
)
```

C++/CLI

```
Curve^ ICreateLine(
    & System.double% RootPoint,
    & System.double% Direction
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

RootPoint

Array containing 3 doubles (x, y, z) for the start point of the line

Direction
Array containing 3 doubles (x, y, z) for the end point of the line

Return Value

Newly created [line](#)

Visual Basic for Applications (VBA) Syntax

See [Modeler::ICreateLine](#).

Remarks

If your application uses [ISurface::CreateTrimmedSheet4](#) or [ISurface::ICreateTrimmedSheet4](#), then your application must also use [ICurve::CreateTrimmedCurve2](#) for the curves created by [IModeler::CreateArc](#) or [IModeler::ICreateArc](#) and [IModeler::CreateLine](#) or [IModeler::ICreateLine](#).

See Also

[IModeler Interface](#)
[IModeler Members](#)
[IModeler::CreateEllipse Method](#)
[IModeler::ICreateEllipse Method](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

ICreateLoftSurface Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : ICreateLoftSurface Method (IModeler)

Creates a loft surface.

.NET Syntax

Visual Basic (Declaration)

```
Function ICreateLoftSurface( _  
    ByVal NCurves As System.Integer, _  
    ByRef CurveArray As Curve, _  
    ByVal BBlendClosed As System.Boolean, _  
    ByVal BForceCubic As System.Boolean, _  
    ByVal NGuides As System.Integer, _  
    ByRef GuideCrvArray As Curve, _  
    ByVal StartMatchingType As System.Integer, _  
    ByVal EndMatchingType As System.Integer, _  
    ByVal NormalAtStartSection As MathVector, _  
    ByVal NormalAtEndSection As MathVector, _  
    ByVal NStartMatchingFaces As System.Integer, _  
    ByRef StartMatchingFaceList As Face2, _  
    ByVal NEndMatchingFaces As System.Integer, _  
    ByRef EndMatchingFaceList As Face2, _  
    ByVal DegeneratedStart As System.Boolean, _  
    ByVal DegeneratedEnd As System.Boolean, _  
    ByVal StartPointOfStartSection As MathPoint, _  
    ByVal StartPointOfEndSection As MathPoint, _  
    ByVal SectionIndexStart As System.Integer, _  
    ByVal SectionIndexEnd As System.Integer, _  
    ByVal GuideIndexStart As System.Integer, _  
    ByVal GuideIndexEnd As System.Integer _  
) As Surface
```

Visual Basic (Usage)



```
Dim instance As IModeler  
Dim NCurves As System.Integer  
Dim CurveArray As Curve  
Dim BBlendClosed As System.Boolean  
Dim BForceCubic As System.Boolean  
Dim NGuides As System.Integer  
Dim GuideCrvArray As Curve  
Dim StartMatchingType As System.Integer
```

```

Dim EndMatchingType As System.Integer
Dim NormalAtStartSection As MathVector
Dim NormalAtEndSection As MathVector
Dim NStartMatchingFaces As System.Integer
Dim StartMatchingFaceList As Face2
Dim NEndMatchingFaces As System.Integer
Dim EndMatchingFaceList As Face2
Dim DegeneratedStart As System.Boolean
Dim DegeneratedEnd As System.Boolean
Dim StartPointOfStartSection As MathPoint
Dim StartPointOfEndSection As MathPoint
Dim SectionIndexStart As System.Integer
Dim SectionIndexEnd As System.Integer
Dim GuideIndexStart As System.Integer
Dim GuideIndexEnd As System.Integer
Dim value As Surface

```

```

value = instance.ICreateLoftSurface(NCurves, CurveArray, BBlendClosed, BForceCubic, NGuides,
GuideCrvArray, StartMatchingType, EndMatchingType, NormalAtStartSection, NormalAtEndSection,
NStartMatchingFaces, StartMatchingFaceList, NEndMatchingFaces, EndMatchingFaceList,
DegeneratedStart, DegeneratedEnd, StartPointOfStartSection, StartPointOfEndSection,
SectionIndexStart, SectionIndexEnd, GuideIndexStart, GuideIndexEnd)

```

C#

```

Surface ICreateLoftSurface(
    System.int NCurves,
    ref Curve CurveArray,
    System.bool BBlendClosed,
    System.bool BForceCubic,
    System.int NGuides,
    ref Curve GuideCrvArray,
    System.int StartMatchingType,
    System.int EndMatchingType,
    MathVector NormalAtStartSection,
    MathVector NormalAtEndSection,
    System.int NStartMatchingFaces,
    ref Face2 StartMatchingFaceList,
    System.int NEndMatchingFaces,
    ref Face2 EndMatchingFaceList,
    System.bool DegeneratedStart,
    System.bool DegeneratedEnd,
    MathPoint StartPointOfStartSection,
    MathPoint StartPointOfEndSection,
    System.int SectionIndexStart,
    System.int SectionIndexEnd,
    System.int GuideIndexStart,
    System.int GuideIndexEnd
)

```

C++/CLI

```

Surface^ ICreateLoftSurface(
    & System.int NCurves,
    & Curve^% CurveArray,
    & System.bool BBlendClosed,
    & System.bool BForceCubic,
    & System.int NGuides,
    & Curve^% GuideCrvArray,
    & System.int StartMatchingType,
    & System.int EndMatchingType,
    & MathVector^ NormalAtStartSection,
    & MathVector^ NormalAtEndSection,
    & System.int NStartMatchingFaces,
    & Face2^% StartMatchingFaceList,
    & System.int NEndMatchingFaces,
    & Face2^% EndMatchingFaceList,
    & System.bool DegeneratedStart,
    & System.bool DegeneratedEnd,
    & MathPoint^ StartPointOfStartSection,

```

```

& MathPoint^ StartPointOfEndSection,
& System.int SectionIndexStart,
& System.int SectionIndexEnd,
& System.int GuideIndexStart,
& System.int GuideIndexEnd
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NCurves

Number of B-spline curves

CurveArray

Array of b-spline [curves](#)

BBlendClosed

True if blend closed, false if not

BForceCubic

True if force surface is cubic, false if not

NGuides

Number of guide curves

GuideCrvArray

Array of guide [curves](#)

StartMatchingType

Start matching type:

- 0 = MATCH_NONE (default)
- 1 = MATCH_NORMAL
- 2 = MATCH_VECTOR
- 3 = MATCH_ALL_FACES or MATCH_FACE_G1
- 4 = MATCH_FACE_G2

EndMatchingType

End matching type:

- 0 = MATCH_NONE (default)
- 1 = MATCH_NORMAL
- 2 = MATCH_VECTOR
- 3 = MATCH_ALL_FACES or MATCH_FACE_G1
- 4 = MATCH_FACE_G2

NormalAtStartSection

[Normal at start section](#) when StartMatchingType is MATCH_NORMAL or MATCH_VECTOR; otherwise, can be NULL

NormalAtEndSection

Array of matching [faces](#) at end section when startMatchingType is MATCH_ALL_FACES, MATCH_FACE_G1, or MATCH_FACE_G2; otherwise, can be NULL

NStartMatchingFaces

Number of matching faces at start section when startMatchingType is MATCH_ALL_FACES, MATCH_FACE_G1, or MATCH_FACE_G2; otherwise, can be 0

StartMatchingFaceList

Array of matching [faces](#) at start section when startMatchingType is MATCH_ALL_FACES, MATCH_FACE_G1, or MATCH_FACE_G2; otherwise, can be NULL

NEndMatchingFaces

Number of matching faces at end section when startMatchingType is MATCH_ALL_FACES, MATCH_FACE_G1, or MATCH_FACE_G2; otherwise, can be 0

EndMatchingFaceList

Array of matching [faces](#) at end section when startMatchingType is MATCH_ALL_FACES, MATCH_FACE_G1, or MATCH_FACE_G2; otherwise, can be NULL

DegeneratedStart

True to degenerate at start, false to not

DegeneratedEnd

True to degenerate at end, false to not

StartPointOfStartSection

[Start point](#) of start section

StartPointOfEndSection

[Start point](#) of end section

SectionIndexStart

Index of start section; default is -1

SectionIndexEnd

Index of end section; default is -1

GuideIndexStart

Index of start guide curve; default is -1

GuideIndexEnd

Index of end guide curve; default is -1

Visual Basic for Applications (VBA) Syntax

See [Modeler::ICreateLoftSurface](#).

See Also

[IModeler Interface](#)

[IModeler Members](#)

[IModeler::CreateLoftSurface Method](#)

[IModeler::CreateBspineSurface Method](#)

[IModeler::CreateConicalSurface2 Method](#)

[IModeler::CreateCoonsBSurface Method](#)

[IModeler::CreateCylindricalSurface2 Method](#)

[IModeler::CreateExtrusionSurface Method](#)

[IModeler::CreateOffsetSurface Method](#)

[IModeler::CreatePlanarSurface2 Method](#)

[IModeler::CreateRuledSurface Method](#)

[IModeler::CreateSphericalSurface2 Method](#)
[IModeler::CreateSweptSurface Method](#)
[IModeler::CreateToroidalSurface Method](#)
[IModeler::ICreateBsplinesSurface Method](#)
[IModeler::ICreateConicalSurface2 Method](#)
[IModeler::ICreateCylindricalSurface2 Method](#)
[IModeler::ICreateExtrusionSurface Method](#)
[IModeler::ICreateOffsetSurface Method](#)
[IModeler::ICreatePlanarSurface2 Method](#)
[IModeler::ICreateRuledSurface Method](#)
[IModeler::ICreateSphericalSurface2 Method](#)
[IModeler::ICreateSweptSurface Method](#)
[IModeler::ICreateToroidalSurface Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

ICreateOffsetSurface Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : ICreateOffsetSurface Method (IModeler)

Creates a surface that is offset from an existing surface.

.NET Syntax

Visual Basic (Declaration)

```
Function ICreateOffsetSurface( _  
    ByVal SurfaceIn As Surface, _  
    ByVal Distance As System.Double _  
) As Surface
```

Visual Basic (Usage)



```
Dim instance As IModeler  
Dim SurfaceIn As Surface  
Dim Distance As System.Double  
Dim value As Surface  
  
value = instance.ICreateOffsetSurface(SurfaceIn, Distance)
```

C#

```
Surface ICreateOffsetSurface(  
    Surface SurfaceIn,  
    System.double Distance  
)
```

C++/CLI

```
Surface^ ICreateOffsetSurface(  
& Surface^ SurfaceIn,  
& System.double Distance  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

SurfaceIn
Offset [surface](#)
Distance
Offset distance

Visual Basic for Applications (VBA) Syntax

See [Modeler::ICreateOffsetSurface](#).

See Also

[IModeler Interface](#)
[IModeler Members](#)
[IModeler::CreateOffsetSurface Method](#)
[IBody2::CreateOffsetSurface Method](#)
[IBody2::ICreateOffsetSurface Method](#)
[IModeler::CreateBsplinesSurface Method](#)
[IModeler::CreateConicalSurface2 Method](#)
[IModeler::CreateCoonsBSurface Method](#)
[IModeler::CreateCylindricalSurface2 Method](#)

[IModeler::CreateExtrusionSurface Method](#)
[IModeler::CreateLoftSurface Method](#)
[IModeler::CreateOffsetSurface Method](#)
[IModeler::CreatePlanarSurface2 Method](#)
[IModeler::CreateRuledSurface Method](#)
[IModeler::CreateSphericalSurface2 Method](#)
[IModeler::CreateSweptSurface Method](#)
[IModeler::CreateToroidalSurface Method](#)
[IModeler::ICreateB splineSurface Method](#)
[IModeler::ICreateConicalSurface2 Method](#)
[IModeler::ICreateCylindricalSurface2 Method](#)
[IModeler::ICreateExtrusionSurface Method](#)
[IModeler::ICreateOffsetSurface Method](#)
[IModeler::ICreatePlanarSurface2 Method](#)
[IModeler::ICreateRuledSurface Method](#)
[IModeler::ICreateSphericalSurface2 Method](#)
[IModeler::ICreateSweptSurface Method](#)
[IModeler::ICreateToroidalSurface Method](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ICreatePCurve Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : ICreatePCurve Method (IModeler)

Creates a pcurve.

.NET Syntax

Visual Basic (Declaration)

```
Function ICreatePCurve( _  
    ByVal Surface As Surface, _  
    ByRef Props As System.Integer, _  
    ByRef Knots As System.Double, _  
    ByRef CtrlPtCoords As System.Double _  
    ) As Curve
```

Visual Basic (Usage)



```
Dim instance As IModeler  
Dim Surface As Surface  
Dim Props As System.Integer  
Dim Knots As System.Double  
Dim CtrlPtCoords As System.Double  
Dim value As Curve  
  
value = instance.ICreatePCurve(Surface, Props, Knots, CtrlPtCoords)
```

C#

```
Curve ICreatePCurve(  
    Surface Surface,  
    ref System.int Props,  
    ref System.double Knots,  
    ref System.double CtrlPtCoords  
)
```

C++/CLI

```
Curve^ ICreatePCurve(  
& Surface^ Surface,  
& System.int% Props,  
& System.double% Knots,  
& System.double% CtrlPtCoords  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

[Surface](#)
[Surface](#) associated with the pcurve
[Props](#)

Array containing 4 integers packed into 2 double elements (see **Remarks**)

Knots

Array of numKnots (see **Remarks**)

CtrlPtCoords

Array of numCtrlPtCoord (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Modeler::ICreatePCurve](#).

Remarks

The Props argument contains the following values:

- DimensionControlPoints
- Order
- NumCtrlPoints
- Periodicity

Length of this array...	Given by...
Knots	numKnots = NumCtrlPoints + Order
CtrlPtCoords	NumCtrlPtCoord = NumCtrlPoints * DimensionControlPoints

See Also

[IModeler Interface](#)
[IModeler Members](#)
[IModeler::CreatePCurve Method](#)
[ICurve::GetPCurveParams Method](#)
[ICurve::IGetPCurveParams Method](#)
[ICurve::IGetPCurveParamsSize Method](#)
[IModeler::CreateBsplineCurve Method](#)
[IModeler::ICreateBsplineCurve Method](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ICreatePlanarSurface Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : ICreatePlanarSurface Method (IModeler)

Obsolete. Superseded by [IModeler::ICreatePlanarSurface2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function ICreatePlanarSurface( _  
    ByRef RootPoint As System.Double, _  
    ByRef Normal As System.Double _  
    ) As Surface
```

Visual Basic (Usage)



```
Dim instance As IModeler  
Dim RootPoint As System.Double  
Dim Normal As System.Double  
Dim value As Surface
```

```
value = instance.ICreatePlanarSurface(RootPoint, Normal)
```

C#

```
Surface ICreatePlanarSurface(  
    ref System.double RootPoint,
```

```
    ref System.double Normal
)
```

C++/CLI

```
Surface^ ICreatePlanarSurface(
&    System.double% RootPoint,
&    System.double% Normal
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

RootPoint

Normal

Visual Basic for Applications (VBA) Syntax

See [Modeler::ICreatePlanarSurface](#).

See Also

[IModeler Interface](#)
[IModeler Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

ICreatePlanarSurface2 Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : ICreatePlanarSurface2 Method (IModeler)

Creates a new infinite planar surface.

.NET Syntax

Visual Basic (Declaration)

```
Function ICreatePlanarSurface2( _
    ByRef RootPoint As System.Double, _
    ByRef Normal As System.Double, _
    ByRef Ref As System.Double _
) As Surface
```

Visual Basic (Usage)



```
Dim instance As IModeler
Dim RootPoint As System.Double
Dim Normal As System.Double
Dim Ref As System.Double
Dim value As Surface
```

```
value = instance.ICreatePlanarSurface2(RootPoint, Normal, Ref)
```

C#

```
Surface ICreatePlanarSurface2(
    ref System.double RootPoint,
    ref System.double Normal,
    ref System.double Ref
)
```

C++/CLI

```
Surface^ ICreatePlanarSurface2(
&    System.double% RootPoint,
&    System.double% Normal,
&    System.double% Ref
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

RootPoint
Array of 3 doubles (x,y,z)
Normal
Array of 3 doubles (x,y,z)
Ref
Array of 3 doubles representing the vector defining the x-axis on the plane

Return Value

Newly created planar [surface](#)

Visual Basic for Applications (VBA) Syntax

See [Modeler::ICreatePlanarSurface2](#).

Remarks

You can use this method with:

- A set of related functions designed to construct a body from trimmed surfaces. It can also be used with trimming curve creation routines to construct a trimmed planar surface.
- Trimming curve creation routines such as [ISurface::AddTrimmingLoop2](#) or [ISurface::!AddTrimmingLoop2](#) to construct a trimmed surface

See [IBody2::CreatePlanarTrimSurfaceDLL](#) or [IBody2::ICreatePlanarTrimSurfaceDLL](#), which allows you to create a planar trimmed surface directly from the vertex points.

See Also

[IModeler Interface](#)
[IModeler Members](#)
[IModeler::CreateBsplinesurface Method](#)
[IModeler::CreateConicalSurface2 Method](#)
[IModeler::CreateCoonsBSurface Method](#)
[IModeler::CreateCylindricalSurface2 Method](#)
[IModeler::CreateExtrusionSurface Method](#)
[IModeler::CreateLoftSurface Method](#)
[IModeler::CreateOffsetSurface Method](#)
[IModeler::CreatePlanarSurface2 Method](#)
[IModeler::CreateRuledSurface Method](#)
[IModeler::CreateSphericalSurface2 Method](#)
[IModeler::CreateSweptSurface Method](#)
[IModeler::CreateToroidalSurface Method](#)
[IModeler::!CreateBsplinesurface Method](#)
[IModeler::!CreateConicalSurface2 Method](#)
[IModeler::!CreateCylindricalSurface2 Method](#)
[IModeler::!CreateExtrusionSurface Method](#)
[IModeler::!CreateLoftSurface Method](#)
[IModeler::!CreateOffsetSurface Method](#)
[IModeler::!CreateRuledSurface Method](#)
[IModeler::!CreateSweptSurface Method](#)
[IModeler::!CreateToroidalSurface Method](#)
[IModeler::!CreateSphericalSurface2 Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ICreateRuledSurface Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : ICreateRuledSurface Method (IModeler)

Creates a ruled surface from the curves.

.NET Syntax

Visual Basic (Declaration)

```
Function ICreateRuledSurface( _  
    ByVal Curve1In As Curve, _  
    ByVal Curve2In As Curve, _  
    ByRef Apex As System.Double _  
    ) As Surface
```

Visual Basic (Usage)



```

Dim instance As IModeler
Dim Curve1In As Curve
Dim Curve2In As Curve
Dim Apex As System.Double
Dim value As Surface

value = instance.ICreateRuledSurface(Curve1In, Curve2In, Apex)

```

C#

```

Surface ICreateRuledSurface(
    Curve Curve1In,
    Curve Curve2In,
    ref System.double Apex
)

```

C++/CLI

```

Surface^ ICreateRuledSurface(
    & Curve^ Curve1In,
    & Curve^ Curve2In,
    & System.double% Apex
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

[Curve1In](#)
 First [curve](#)
[Curve2In](#)
 Second [curve](#)
[Apex](#)
 Array of 3 doubles, (x, y, z), the apex point

Return Value

Newly created ruled [surface](#)

Visual Basic for Applications (VBA) Syntax

See [Modeler::ICreateRuledSurface](#).

Remarks

Valid input curves are b-curves.

See Also

[IModeler Interface](#)
[IModeler Members](#)
[IModeler::CreateBsplinesSurface Method](#)
[IModeler::CreateConicalSurface2 Method](#)
[IModeler::CreateCoonsBSurface Method](#)
[IModeler::CreateCylindricalSurface2 Method](#)
[IModeler::CreateExtrusionSurface Method](#)
[IModeler::CreateLoftSurface Method](#)
[IModeler::CreateOffsetSurface Method](#)
[IModeler::CreatePlanarSurface2 Method](#)
[IModeler::CreateRuledSurface Method](#)
[IModeler::CreateSphericalSurface2 Method](#)
[IModeler::CreateSweptSurface Method](#)
[IModeler::CreateToroidalSurface Method](#)
[IModeler::ICreateBsplinesSurface Method](#)
[IModeler::ICreateConicalSurface2 Method](#)
[IModeler::ICreateCylindricalSurface2 Method](#)
[IModeler::ICreateExtrusionSurface Method](#)
[IModeler::ICreateLoftSurface Method](#)
[IModeler::ICreateOffsetSurface Method](#)
[IModeler::ICreatePlanarSurface2 Method](#)
[IModeler::ICreateSphericalSurface2 Method](#)
[IModeler::ICreateSweptSurface Method](#)
[IModeler::ICreateToroidalSurface Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

ICreateRuledSurfaceFromEdge Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : ICreateRuledSurfaceFromEdge Method (IModeler)

Creates a ruled surface using the specified edges and returns a surface body.

.NET Syntax

Visual Basic (Declaration)

```
Function ICreateRuledSurfaceFromEdge( _
    ByVal ModDoc As ModelDoc2, _
    ByVal NumOfEdges As System.Integer, _
    ByRef Edges As Edge, _
    ByVal Type As System.Integer, _
    ByVal Length As System.Double, _
    ByVal FlipPullDir As System.Boolean, _
    ByVal FlipDir As System.Boolean, _
    ByVal TrimAndSew As System.Boolean, _
    ByVal Angle As System.Double, _
    ByVal CoordInput As System.Boolean, _
    ByVal X As System.Double, _
    ByVal Y As System.Double, _
    ByVal Z As System.Double, _
    ByVal BConnectSurface As System.Boolean, _
    ByRef RuledSurface As Body2 _
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IModeler
Dim ModDoc As ModelDoc2
Dim NumOfEdges As System.Integer
Dim Edges As Edge
Dim Type As System.Integer
Dim Length As System.Double
Dim FlipPullDir As System.Boolean
Dim FlipDir As System.Boolean
Dim TrimAndSew As System.Boolean
Dim Angle As System.Double
Dim CoordInput As System.Boolean
Dim X As System.Double
Dim Y As System.Double
Dim Z As System.Double
Dim BConnectSurface As System.Boolean
Dim RuledSurface As Body2
Dim value As System.Integer
```

```
value = instance.ICreateRuledSurfaceFromEdge(ModDoc, NumOfEdges, Edges, Type, Length, FlipPullDir, FlipDir, TrimAndSew, Angle, CoordInput, X, Y, Z, BConnectSurface, RuledSurface)
```

C#

```
System.int ICreateRuledSurfaceFromEdge(
    ModelDoc2 ModDoc,
    System.int NumOfEdges,
    ref Edge Edges,
    System.int Type,
    System.double Length,
    System.bool FlipPullDir,
    System.bool FlipDir,
    System.bool TrimAndSew,
    System.double Angle,
    System.bool CoordInput,
    System.double X,
    System.double Y,
    System.double Z,
```

```

        System.bool BConnectSurface,
        out Body2 RuledSurface
    )

```

C++/CLI

```

System.int ICreateRuledSurfaceFromEdge(
    & ModelDoc2^ ModDoc,
    & System.int NumOfEdges,
    & Edge^% Edges,
    & System.int Type,
    & System.double Length,
    & System.bool FlipPullDir,
    & System.bool FlipDir,
    & System.bool TrimAndSew,
    & System.double Angle,
    & System.bool CoordInput,
    & System.double X,
    & System.double Y,
    & System.double Z,
    & System.bool BConnectSurface,
    & [Out] Body2^ RuledSurface
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ModDoc

[Model document](#) in which to create surface body

NumOfEdges

Number of edges on which to create ruled surface

Edges

Array of edges on which to create ruled surface

NOTE: Currently only a single edge is supported.

Type

- 0 = Tangent to Surface
- 1 = Normal to Surface
- 2 = Tapered to Vector
- 3 = Perpendicular to Vector
- 4 = Sweep

Length

Distance at which to create the surface; valid for Tangent to Surface, Tapered to Vector, Perpendicular to Vector, and Sweep types only

FlipPullDir

True to flip the pull direction, false to not; valid for Normal to Surface and Tapered to Vector types only

FlipDir

True to flip the direction, false to not; valid for Perpendicular to Vector type only

TrimAndSew

True to trim and knit the surface, false to not

Angle

Angle for Tapered to Vector type only

CoordInput

True to enable coordinate input, false if not; for Sweep type only

X

x coordinate

Y

y coordinate

Z

z coordinate

BConnectSurface

True to remove any connecting surfaces, false to not

RuledSurface

Ruled surface [body](#)

Return Value

0 if ruled surface body created, -1 if not

Remarks

This method returns a surface body, unlike [IFeatureManager::InsertRuledSurfaceFromEdge2](#), which returns a feature.

If you [select](#) an edge with a selection mark of 4, the default face is used. If you select an edge with a selection mark of 6, the alternate face is used.

See Also

[IModeler Interface](#)
[IModeler Members](#)
[IModeler::ICreateRuledSurface Method](#)
[IModeler::CreateRuledSurfaceFromEdge Method](#)
[IModeler::CreateRuledSurface Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ICreateSheetFromFaces Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : ICreateSheetFromFaces Method (IModeler)

Creates a sheet (surface) body from connected faces.

.NET Syntax

Visual Basic (Declaration)

```
Function ICreateSheetFromFaces( _  
    ByVal FaceCount As System.Integer, _  
    ByRef FaceArr As Face2 _  
) As Body2
```

Visual Basic (Usage)



```
Dim instance As IModeler  
Dim FaceCount As System.Integer  
Dim FaceArr As Face2  
Dim value As Body2
```

```
value = instance.ICreateSheetFromFaces(FaceCount, FaceArr)
```

C#

```
Body2 ICreateSheetFromFaces(  
    System.int FaceCount,  
    ref Face2 FaceArr  
)
```

C++/CLI

```
Body2^ ICreateSheetFromFaces(  
& System.int FaceCount,  
& Face2^% FaceArr  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FaceCount
Number of connected faces
FaceArr
Array of connected [faces](#)

Return Value

Newly created [body](#).

Visual Basic for Applications (VBA) Syntax

See [Modeler::ICreateSheetFromFaces](#).

See Also

[IModeler Interface](#)
[IModeler Members](#)
[IModeler::CreateSheetFromSurface Method](#)
[IModeler::CreateSheetFromFaces Method](#)
[IModeler::ICreateSheetFromSurface2 Method](#)
[IFace2::CreateSheetBody Method](#)
[IFace2::CreateSheetBodyByFaceExtension Method](#)
[IFace2::ICreateSheetBody Method](#)
[IFace2::ICreateSheetBodyByFaceExtension Method](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ICreateSheetFromSurface Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : ICreateSheetFromSurface Method (IModeler)

Obsolete. Superseded by [IModeler::ICreateSheetFromSurface2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function ICreateSheetFromSurface( _  
    ByVal SurfaceIn As Surface, _  
    ByRef UvRange As System.Double _  
) As Body
```

Visual Basic (Usage)



```
Dim instance As IModeler  
Dim SurfaceIn As Surface  
Dim UvRange As System.Double  
Dim value As Body  
  
value = instance.ICreateSheetFromSurface(SurfaceIn, UvRange)
```

C#

```
Body ICreateSheetFromSurface(  
    Surface SurfaceIn,  
    ref System.double UvRange  
)
```

C++/CLI

```
Body^ ICreateSheetFromSurface(  
& Surface^ SurfaceIn,  
& System.double% UvRange  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

SurfaceIn

UvRange

Visual Basic for Applications (VBA) Syntax

See [Modeler::ICreateSheetFromSurface](#).

See Also

[IModeler Interface](#)
[IModeler Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

ICreateSheetFromSurface2 Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : ICreateSheetFromSurface2 Method (IModeler)

Creates a sheet (surface) body from a surface.

.NET Syntax

Visual Basic (Declaration)

```
Function ICreateSheetFromSurface2( _  
    ByVal SurfaceIn As Surface, _  
    ByRef UvRange As System.Double _  
    ) As Body2
```

Visual Basic (Usage)



```
Dim instance As IModeler  
Dim SurfaceIn As Surface  
Dim UvRange As System.Double  
Dim value As Body2
```

```
value = instance.ICreateSheetFromSurface2(SurfaceIn, UvRange)
```

C#

```
Body2 ICreateSheetFromSurface2(  
    Surface SurfaceIn,  
    ref System.double UvRange  
)
```

C++/CLI

```
Body2^ ICreateSheetFromSurface2(  
& Surface^ SurfaceIn,  
& System.double% UvRange  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

SurfaceIn
[Surface](#) from which you want to create this sheet body
UvRange
Array of UV values for this surface

Return Value

Newly created [body](#)

Visual Basic for Applications (VBA) Syntax

See [Modeler::ICreateSheetFromSurface2](#).

See Also

[IModeler Interface](#)
[IModeler Members](#)
[IModeler::CreateSheetFromSurface Method](#)
[IModeler::ICreateSheetFromFaces Method](#)
[ISurface::CreateTrimmedSheet Method](#)
[IModeler::CreateSheetFromFaces Method](#)

Availability

ICreateSphericalSurface Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : ICreateSphericalSurface Method (IModeler)

Obsolete. Superseded by [IModeler::ICreateSphericalSurface2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function ICreateSphericalSurface( _  
    ByRef Center As System.Double, _  
    ByVal Radius As System.Double _  
) As Surface
```

Visual Basic (Usage)



```
Dim instance As IModeler  
Dim Center As System.Double  
Dim Radius As System.Double  
Dim value As Surface
```

```
value = instance.ICreateSphericalSurface(Center, Radius)
```

C#

```
Surface ICreateSphericalSurface(  
    ref System.double Center,  
    System.double Radius  
)
```

C++/CLI

```
Surface^ ICreateSphericalSurface(  
& System.double% Center,  
& System.double Radius  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Center

Radius

Visual Basic for Applications (VBA) Syntax

See [Modeler::ICreateSphericalSurface](#).

See Also

[IModeler Interface](#)
[IModeler Members](#)

ICreateSphericalSurface2 Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : ICreateSphericalSurface2 Method (IModeler)

Creates an untrimmed spherical surface.

.NET Syntax

Visual Basic (Declaration)

```
Function ICreateSphericalSurface2( _
    ByRef Center As System.Double, _
    ByRef Axis As System.Double, _
    ByRef RefDir As System.Double, _
    ByVal Radius As System.Double _
) As Surface
```

Visual Basic (Usage)



```
Dim instance As IModeler
Dim Center As System.Double
Dim Axis As System.Double
Dim RefDir As System.Double
Dim Radius As System.Double
Dim value As Surface

value = instance.ICreateSphericalSurface2(Center, Axis, RefDir, Radius)
```

C#

```
Surface ICreateSphericalSurface2(
    ref System.double Center,
    ref System.double Axis,
    ref System.double RefDir,
    System.double Radius
)
```

C++/CLI

```
Surface^ ICreateSphericalSurface2(
    & System.double% Center,
    & System.double% Axis,
    & System.double% RefDir,
    & System.double Radius
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Center
Array containing 3 doubles for XYZ location of the center of the spherical surface

Axis
Array containing 3 doubles

RefDir
Array containing 3 doubles

Radius
Radius at the center

Return Value

Untrimmed spherical [surface](#)

Visual Basic for Applications (VBA) Syntax

See [Modeler::ICreateSphericalSurface2](#).

Remarks

The axis argument is the axis of the sphere; the refDir argument is perpendicular to the axis and through the center of the sphere and defines the origin of the UV space.

You can trim the resulting surface to generate a sheet body, for example, using [ISurface::CreateTrimmedSheet](#).

See Also

[IModeler Interface](#)
[IModeler Members](#)
[IModeler::CreateBsplinesSurface Method](#)
[IModeler::CreateConicalSurface2 Method](#)
[IModeler::CreateCoonsBSurface Method](#)
[IModeler::CreateCylindricalSurface2 Method](#)
[IModeler::CreateExtrusionSurface Method](#)

[IModeler::CreateLoftSurface Method](#)
[IModeler::CreateOffsetSurface Method](#)
[IModeler::CreatePlanarSurface2 Method](#)
[IModeler::CreateRuledSurface Method](#)
[IModeler::CreateSphericalSurface2 Method](#)
[IModeler::CreateSweptSurface Method](#)
[IModeler::CreateToroidalSurface Method](#)
[IModeler::ICreateB splineSurface Method](#)
[IModeler::ICreateConicalSurface2 Method](#)
[IModeler::ICreateCylindricalSurface2 Method](#)
[IModeler::ICreateExtrusionSurface Method](#)
[IModeler::ICreateLoftSurface Method](#)
[IModeler::ICreateOffsetSurface Method](#)
[IModeler::ICreatePlanarSurface2 Method](#)
[IModeler::ICreateSweptSurface Method](#)
[IModeler::ICreateToroidalSurface Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ICreateSweptSurface Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : ICreateSweptSurface Method (IModeler)

Creates a swept surface from a curve.

.NET Syntax

Visual Basic (Declaration)

```
Function ICreateSweptSurface( _  
    ByVal CurveIn As Curve, _  
    ByRef Dir As System.Double _  
) As Surface
```

Visual Basic (Usage)



```
Dim instance As IModeler  
Dim CurveIn As Curve  
Dim Dir As System.Double  
Dim value As Surface  
  
value = instance.ICreateSweptSurface(CurveIn, Dir)
```

C#

```
Surface ICreateSweptSurface(  
    Curve CurveIn,  
    ref System.double Dir  
)
```

C++/CLI

```
Surface^ ICreateSweptSurface(  
& Curve^ CurveIn,  
& System.double% Dir  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

CurveIn

[Curve](#) to use to create the swept surface

Dir

3 doubles describing the direction vector to sweep the curve along (see **Remarks**)

Return Value

Newly created [surface](#)

Visual Basic for Applications (VBA) Syntax

See [Modeler::ICreateSweptSurface](#).

Remarks

This method creates an infinite surface. To create a trimmed surface, add a trimming loop with [ISurface::AddTrimmingLoop2](#) or [ISurface::!AddTrimmingLoop2](#).

The Dir array is:

- Dir[0] = X component of the direction vector
- Dir[1] = Y component of the direction vector
- Dir[2] = Z component of the direction vector

See Also

[IModeler Interface](#)
[IModeler Members](#)
[IModeler::CreateBsplinesSurface Method](#)
[IModeler::CreateConicalSurface2 Method](#)
[IModeler::CreateCoonsBSurface Method](#)
[IModeler::CreateCylindricalSurface2 Method](#)
[IModeler::CreateExtrusionSurface Method](#)
[IModeler::CreateLoftSurface Method](#)
[IModeler::CreateOffsetSurface Method](#)
[IModeler::CreatePlanarSurface2 Method](#)
[IModeler::CreateRuledSurface Method](#)
[IModeler::CreateSphericalSurface2 Method](#)
[IModeler::CreateToroidalSurface Method](#)
[IModeler::!CreateBsplinesSurface Method](#)
[IModeler::!CreateConicalSurface2 Method](#)
[IModeler::!CreateCylindricalSurface2 Method](#)
[IModeler::!CreateCylindricalSurface2 Method](#)
[IModeler::!CreateLoftSurface Method](#)
[IModeler::!CreateOffsetSurface Method](#)
[IModeler::!CreatePlanarSurface2 Method](#)
[IModeler::!CreateRuledSurface Method](#)
[IModeler::!CreateSheetFromSurface2 Method](#)
[IModeler::!CreateSphericalSurface2 Method](#)
[IModeler::!CreateToroidalSurface Method](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ICreateToroidalSurface Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : ICreateToroidalSurface Method (IModeler)

Creates an untrimmed toroidal surface from the specified arguments.

.NET Syntax

Visual Basic (Declaration)

```
Function ICreateToroidalSurface( _  
    ByRef Center As System.Double, _  
    ByRef Axis As System.Double, _  
    ByRef RefDirection As System.Double, _  
    ByVal MajorRadius As System.Double, _  
    ByVal MinorRadius As System.Double _  
) As Surface
```

Visual Basic (Usage)



```
Dim instance As IModeler  
Dim Center As System.Double  
Dim Axis As System.Double  
Dim RefDirection As System.Double  
Dim MajorRadius As System.Double  
Dim MinorRadius As System.Double  
Dim value As Surface
```

```
value = instance.ICreateToroidalSurface(Center, Axis, RefDirection, MajorRadius, MinorRadius)
```

C#

```
Surface ICreateToroidalSurface(
    ref System.double Center,
    ref System.double Axis,
    ref System.double RefDirection,
    System.double MajorRadius,
    System.double MinorRadius
)
```

C++/CLI

```
Surface^ ICreateToroidalSurface(
    & System.double% Center,
    & System.double% Axis,
    & System.double% RefDirection,
    & System.double MajorRadius,
    & System.double MinorRadius
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Center

Array containing 3 doubles (see **Remarks**)

Axis

Array containing 3 doubles (see **Remarks**)

RefDirection

Array containing 3 doubles (see **Remarks**)

MajorRadius

See **Remarks**

MinorRadius

See **Remarks**

Visual Basic for Applications (VBA) Syntax

See [Modeler::ICreateToroidalSurface](#).

Remarks

Input arguments:

- Center XYZ location that represents the center of the spherical surface.
- Axis XYZ direction of the axis of the spherical surface.
- RefDirection XYZ direction of the reference axis of the spherical surface; it must be perpendicular to the axis.
- MajorRadius major radius of the toroidal surface.
- MinorRadius minor radius of the toroidal surface.

The resulting surface can be trimmed using a method like [ISurface::CreateTrimmedSheet](#) to generate a sheet body.

See Also

[IModeler Interface](#)

[IModeler Members](#)

[IModeler::CreateBspineSurface Method](#)

[IModeler::CreateConicalSurface2 Method](#)

[IModeler::CreateCoonsBSurface Method](#)

[IModeler::CreateCylindricalSurface2 Method](#)

[IModeler::CreateExtrusionSurface Method](#)

[IModeler::CreateLoftSurface Method](#)

[IModeler::CreateOffsetSurface Method](#)

[IModeler::CreatePlanarSurface2 Method](#)

[IModeler::CreateRuledSurface Method](#)

[IModeler::CreateSphericalSurface2 Method](#)

[IModeler::CreateSweptSurface Method](#)

[IModeler::CreateToroidalSurface Method](#)

[IModeler::ICreateBspineSurface Method](#)

[IModeler::ICreateConicalSurface2 Method](#)

[IModeler::ICreateCylindricalSurface2 Method](#)

[IModeler::ICreateExtrusionSurface Method](#)

[IModeler::ICreateLoftSurface Method](#)

[IModeler::ICreateOffsetSurface Method](#)

[IModeler::ICreatePlanarSurface2 Method](#)

[IModeler::ICreateRuledSurface Method](#)

[IModeler::ICreateSweptSurface Method](#)

Availability

ICreateWireBody Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : ICreateWireBody Method (IModeler)

Creates a wire body from the input entities, which can be edges or curves.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function ICreateWireBody( _
    ByVal Count As System.Integer, _
    ByRef EntArr As System.Object, _
    ByVal Option As System.Integer _
) As Body2
```

Visual Basic (Usage)



```
Dim instance As IModeler
Dim Count As System.Integer
Dim EntArr As System.Object
Dim Option As System.Integer
Dim value As Body2
```

```
value = instance.ICreateWireBody(Count, EntArr, Option)
```

C#

```
Body2 ICreateWireBody(
    System.int Count,
    ref System.object EntArr,
    System.int Option
)
```

C++/CLI

```
Body2^ ICreateWireBody(
    & System.int Count,
    & System.Object^% EntArr,
    & System.int Option
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Count

Number of input entities ([curves](#) or [edges](#))

EntArr

Array of input entities ([curves](#) or [edges](#))

Option

Option as defined in [swCreateWireBodyOptions_e](#) (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Modeler::ICreateWireBody](#).

Remarks

Wire bodies contain wires, loops, coedges, edges, and vertices. Wires typically represent profiles, construction lines, and center lines of swept shapes. Wires can also represent wire frames that, when surfaced, form shells.

The swCreateWireBodyMergeCurves option only works for curves with position and tangent continuity.

See Also

[IModeler Interface](#)
[IModeler Members](#)
[IModeler::CreateWireBody Method](#)
[ICurve::CreateWireBody Method](#)
[IEdge::CreateWireBody Method](#)
[IImportedCurveFeatureData::SetBody Method](#)
[ISketchSegment::CreateWireBody Method](#)
[IModeler::ICreateBodiesFromSheets2 Method](#)
[IModeler::ICreateBodyFromBox Method](#)
[IModeler::ICreateBodyFromCone Method](#)
[IModeler::ICreateBodyFromCyl Method](#)
[IModeler::ICreateBodyFromFaces2 Method](#)
[IModeler::ICreateBrepBody3 Method](#)
[IModeler::ICreateExtrudedBody Method](#)
[IModeler::ICreateSweptBody Method](#)
[IModeler::ICreateBodiesFromSheets2 Method](#)
[IModeler::ICreateBodyFromBox2 Method](#)
[IModeler::ICreateBodyFromCone2 Method](#)
[IModeler::ICreateBodyFromCyl2 Method](#)
[IModeler::ICreateBodyFromFaces3 Method](#)
[IModeler::ICreateBrepBody3 Method](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

IDeleteFacesFromSheetBody Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : IDeleteFacesFromSheetBody Method (IModeler)

Deletes the unused faces of the sheet body.

I.NET Syntax

Visual Basic (Declaration)

```
Function IDeleteFacesFromSheetBody( _  
    ByVal Count As System.Integer, _  
    ByRef FaceVar As Face2 _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModeler  
Dim Count As System.Integer  
Dim FaceVar As Face2  
Dim value As System.Boolean
```

```
value = instance.IDeleteFacesFromSheetBody(Count, FaceVar)
```

C#

```
System.bool IDeleteFacesFromSheetBody(  
    System.int Count,  
    ref Face2 FaceVar  
)
```

C++/CLI

```
System.bool IDeleteFacesFromSheetBody(  
& System.int Count,  
& Face2^% FaceVar  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Count

Number of faces to delete

FaceVar

Array of [faces](#) to delete from the sheet (surface) body

Return Value

True if all of the faces are deleted, false if not

Visual Basic for Applications (VBA) Syntax

See [Modeler::IDeleteFacesFromSheetBody](#).

Remarks

Use this method to remove the unused faces from the sheet body created by [IModeler::CreateBrepBody3](#) and [IModeler::ICreateBrepBody3](#).

See Also

[IModeler Interface](#)
[IModeler Members](#)
[IModeler::DeleteFacesFromSheetBody Method](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

IFindTwoEdgeMaxDeviation Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : IFindTwoEdgeMaxDeviation Method (IModeler)

Finds the maximum distance between two edges.

.NET Syntax

Visual Basic (Declaration)

```
Function IFindTwoEdgeMaxDeviation( _  
    ByVal PEdge1 As Edge, _  
    ByVal PEdge2 As Edge _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IModeler  
Dim PEdge1 As Edge  
Dim PEdge2 As Edge  
Dim value As System.Double
```

```
value = instance.IFindTwoEdgeMaxDeviation(PEdge1, PEdge2)
```

C#

```
System.double IFindTwoEdgeMaxDeviation(  
    Edge PEdge1,  
    Edge PEdge2  
)
```

C++/CLI

```
System.double IFindTwoEdgeMaxDeviation(  
& Edge^ PEdge1,  
& Edge^ PEdge2  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PEdge1

First [edge](#)

PEdge2

Second [edge](#)

Return Value

Maximum distance between the two edges

Visual Basic for Applications (VBA) Syntax

See [Modeler::IFindTwoEdgeMaxDeviation](#).

See Also

[IModeler Interface](#)
[IModeler Members](#)
[IModeler::FindTwoEdgeMaxDeviation Method](#)

Availability

SOLIDWORKS 99, datecode 1999207

SOLIDWORKS API Help

[Send comments on this topic.](#)

ImprintingFaces Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : ImprintingFaces Method (IModeler)

Imprints the specified tool faces onto the specified target faces.

.NET Syntax

Visual Basic (Declaration)

```
Sub ImprintingFaces( _  
    ByRef TargetEdges As Edge, _  
    ByRef ToolEdges As Edge, _  
    ByRef TargetVertices As Vertex, _  
    ByRef ToolVertices As Vertex _  
)
```

Visual Basic (Usage)



```
Dim instance As IModeler  
Dim TargetEdges As Edge  
Dim ToolEdges As Edge  
Dim TargetVertices As Vertex  
Dim ToolVertices As Vertex  
  
instance.ImprintingFaces(TargetEdges, ToolEdges, TargetVertices, ToolVertices)
```

C#

```
void ImprintingFaces(  
    out Edge TargetEdges,  
    out Edge ToolEdges,  
    out Vertex TargetVertices,  
    out Vertex ToolVertices  
)
```

C++/CLI

```
void ImprintingFaces(  
& [Out] Edge^ TargetEdges,  
& [Out] Edge^ ToolEdges,  
& [Out] Vertex^ TargetVertices,  
& [Out] Vertex^ ToolVertices  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

TargetEdges
Array of target [edges](#)
ToolEdges
Array of tool [edges](#)
TargetVertices
Array of target [vertices](#)
ToolVertices

Array of tool [vertices](#)

Visual Basic for Applications (VBA) Syntax

See [Modeler::ImprintingFaces](#).

Remarks

Call [IModeler::ImprintingFacesCount2](#) before calling this method.

The target and tool faces must:

- belong to different bodies.
- intersect each other.

See Also

[IModeler Interface](#)
[IModeler Members](#)
[IModeler::ImprintingFaces Method](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

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ImprintingFacesCount Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : ImprintingFacesCount Method (IModeler)

Obsolete. Superseded by [IModeler::ImprintingFacesCount2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IImprintingFacesCount(  
    ByVal NTargetFaces As System.Integer, _  
    ByRef TargetFaceArray As Face, _  
    ByVal NToolFaces As System.Integer, _  
    ByRef ToolFaceArray As Face, _  
    ByVal Options As System.Integer, _  
    ByRef NTargetEdges As System.Integer, _  
    ByRef NtoolEdges As System.Integer, _  
    ByRef NtargetVertices As System.Integer, _  
    ByRef ToolVertices As System.Integer _  
) As System.Boolean
```

Visual Basic (Usage)

```
Dim instance As IModeler  
Dim NTargetFaces As System.Integer  
Dim TargetFaceArray As Face  
Dim NToolFaces As System.Integer  
Dim ToolFaceArray As Face  
Dim Options As System.Integer  
Dim NTargetEdges As System.Integer  
Dim NtoolEdges As System.Integer  
Dim NtargetVertices As System.Integer  
Dim ToolVertices As System.Integer  
Dim value As System.Boolean
```

```
value = instance.IImprintingFacesCount(NTargetFaces, TargetFaceArray, NToolFaces, ToolFaceArray,  
Options, NTargetEdges, NtoolEdges, NtargetVertices, ToolVertices)
```

C#



```

System.bool IImprintingFacesCount(
    System.int NTargetFaces,
    ref Face TargetFaceArray,
    System.int NToolFaces,
    ref Face ToolFaceArray,
    System.int Options,
    out System.int NTargetEdges,
    out System.int NtoolEdges,
    out System.int NtargetVertices,
    out System.int ToolVertices
)

```

C++/CLI

```

System.bool IImprintingFacesCount(
    & System.int NTargetFaces,
    & Face^% TargetFaceArray,
    & System.int NToolFaces,
    & Face^% ToolFaceArray,
    & System.int Options,
    & [Out] System.int NTargetEdges,
    & [Out] System.int NtoolEdges,
    & [Out] System.int NtargetVertices,
    & [Out] System.int ToolVertices
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NTargetFaces

TargetFaceArray

NToolFaces

ToolFaceArray

Options

NTargetEdges

NtoolEdges

NtargetVertices

ToolVertices

Visual Basic for Applications (VBA) Syntax

See [Modeler::IImprintingFacesCount](#).

See Also

[IModeler Interface](#)

[IModeler Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IImprintingFacesCount2 Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : IImprintingFacesCount2 Method (IModeler)

Gets the number of imprinted edges and vertices in the model.

.NET Syntax

Visual Basic (Declaration)

```

Function IImprintingFacesCount2( _
    ByVal NTargetFaces As System.Integer, _
    ByRef TargetFaceArray As Face2, _
    ByVal NToolFaces As System.Integer, _
    ByRef ToolFaceArray As Face2, _

```

```

    ByVal Options As System.Integer, _
    ByRef NTargetEdges As System.Integer, _
    ByRef NtoolEdges As System.Integer, _
    ByRef NtargetVertices As System.Integer, _
    ByRef ToolVertices As System.Integer _
) As System.Boolean

```

Visual Basic (Usage)



```

Dim instance As IModeler
Dim NTargetFaces As System.Integer
Dim TargetFaceArray As Face2
Dim NToolFaces As System.Integer
Dim ToolFaceArray As Face2
Dim Options As System.Integer
Dim NTargetEdges As System.Integer
Dim NtoolEdges As System.Integer
Dim NtargetVertices As System.Integer
Dim ToolVertices As System.Integer
Dim value As System.Boolean

```

```

value = instance.IImprintingFacesCount2(NTargetFaces, TargetFaceArray, NToolFaces, ToolFaceArray,
Options, NTargetEdges, NtoolEdges, NtargetVertices, ToolVertices)

```

C#

```

System.bool IImprintingFacesCount2(
    System.int NTargetFaces,
    ref Face2 TargetFaceArray,
    System.int NToolFaces,
    ref Face2 ToolFaceArray,
    System.int Options,
    out System.int NTargetEdges,
    out System.int NtoolEdges,
    out System.int NtargetVertices,
    out System.int ToolVertices
)

```

C++/CLI

```

System.bool IImprintingFacesCount2(
    & System.int NTargetFaces,
    & Face2^% TargetFaceArray,
    & System.int NToolFaces,
    & Face2^% ToolFaceArray,
    & System.int Options,
    & [Out] System.int NTargetEdges,
    & [Out] System.int NtoolEdges,
    & [Out] System.int NtargetVertices,
    & [Out] System.int ToolVertices
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NTargetFaces

Number of faces in the target body

TargetFaceArray

Array of the [faces](#) that describe the target body

NToolFaces

Number of faces in the tool body

ToolFaceArray

Array of the [faces](#) that describe the tool body

Options

Options for this operation as defined in [swImprintingFacesOpts_e](#)

NTargetEdges

Number of edges returned from this operation

NtoolEdges

Number of tool edges returned from this operation

NtargetVertices

Number of target vertices returned from this operation

ToolVertices

Number of tool vertices returned from this operation

Return Value

True if the operation is successful, false if not

Visual Basic for Applications (VBA) Syntax

See [Modeler::ImprintingFacesCount2](#).

Remarks

Call this method before calling [IModeler::ImprintingFaces](#).

See Also

[IModeler Interface](#)

[IModeler Members](#)

[IModeler::ImprintingFaces Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IMakeManifoldBodies Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : IMakeManifoldBodies Method (IModeler)

Creates manifold bodies from the specified non-manifold body.

.NET Syntax

Visual Basic (Declaration)

```
Function IMakeManifoldBodies( _  
    ByVal NonManifoldIn As Body2, _  
    ByVal Count As System.Integer _  
) As Body2
```

Visual Basic (Usage)



```
Dim instance As IModeler  
Dim NonManifoldIn As Body2  
Dim Count As System.Integer  
Dim value As Body2  
  
value = instance.IMakeManifoldBodies(NonManifoldIn, Count)
```

C#

```
Body2 IMakeManifoldBodies(  
    Body2 NonManifoldIn,  
    System.int Count  
)
```

C++/CLI

```
Body2^ IMakeManifoldBodies(  
& Body2^ NonManifoldIn,  
& System.int Count  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NonManifoldIn
Non-manifold [bodies](#)
Count
Number of manifold bodies

Return Value

[Manifold bodies](#)

Visual Basic for Applications (VBA) Syntax

See [Modeler::IMakeManifoldBodies](#).

Remarks

Call [IModeler::GetManifoldBodiesCount](#) before calling this method to determine the size of the array for that method.

See Also

[IModeler Interface](#)
[IModeler Members](#)
[IModeler::MakeManifoldBodies Method](#)
[IModeler::GeneralTopology Property](#)

Availability

SOLIDWORKS 2006 FCS, Revision Number 14.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

IMergeCurves Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : IMergeCurves Method (IModeler)

Merges multiple curves into one curve.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function IMergeCurves( _  
    ByVal CurveCount As System.Integer, _  
    ByRef CurveArr As Curve _  
) As Curve
```

Visual Basic (Usage)



```
Dim instance As IModeler  
Dim CurveCount As System.Integer  
Dim CurveArr As Curve  
Dim value As Curve  
  
value = instance.IMergeCurves(CurveCount, CurveArr)
```

C#

```
Curve IMergeCurves(  
    System.int CurveCount,  
    ref Curve CurveArr  
)
```

C++/CLI

```
Curve^ IMergeCurves(  
& System.int CurveCount,  
& Curve^% CurveArr  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

CurveCount
Number of curves to merge

CurveArr
Array of [curves](#) to merge

Return Value

Newly created merged [curve](#) or NULL if merge fails

Visual Basic for Applications (VBA) Syntax

See [Modeler::IMergeCurves](#).

See Also

[IModeler Interface](#)
[IModeler Members](#)
[IModeler::MergeCurves Method](#)

Availability

SOLIDWORKS 2006 FCS, Revision Number 14.0

SOLIDWORKS API Help

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ImprintingFaces Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : ImprintingFaces Method (IModeler)

Imprints the specified tool faces onto the specified target faces.

.NET Syntax

Visual Basic (Declaration)

```
Function ImprintingFaces( _  
    ByVal TargetFaceArray As System.Object, _  
    ByVal ToolFaceArray As System.Object, _  
    ByVal Options As System.Integer, _  
    ByRef TargetEdges As System.Object, _  
    ByRef ToolEdges As System.Object, _  
    ByRef TargetVertices As System.Object, _  
    ByRef ToolVertices As System.Object _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModeler  
Dim TargetFaceArray As System.Object  
Dim ToolFaceArray As System.Object  
Dim Options As System.Integer  
Dim TargetEdges As System.Object  
Dim ToolEdges As System.Object  
Dim TargetVertices As System.Object  
Dim ToolVertices As System.Object  
Dim value As System.Boolean
```

```
value = instance.ImprintingFaces(TargetFaceArray, ToolFaceArray, Options, TargetEdges, ToolEdges,  
TargetVertices, ToolVertices)
```

C#

```
System.bool ImprintingFaces(  
    System.object TargetFaceArray,  
    System.object ToolFaceArray,  
    System.int Options,  
    out System.object TargetEdges,  
    out System.object ToolEdges,  
    out System.object TargetVertices,
```

```
        out System.Object ToolVertices
    )
```

C++/CLI

```
System.bool ImprintingFaces(
    & System.Object^ TargetFaceArray,
    & System.Object^ ToolFaceArray,
    & System.int Options,
    & [Out] System.Object^ TargetEdges,
    & [Out] System.Object^ ToolEdges,
    & [Out] System.Object^ TargetVertices,
    & [Out] System.Object^ ToolVertices
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

TargetFaceArray
Array of [faces](#) that describe the target body
ToolFaceArray
Array of [faces](#) that describe the tool body
Options
Options for this operation as defined in [swImprintingFacesOpts_e](#)
TargetEdges
Array target [edges](#)
ToolEdges
Array of tool [edges](#)
TargetVertices
Array of target [vertices](#)
ToolVertices
Array of tool [vertices](#)

Return Value

True if the operation was successful, false if not

Visual Basic for Applications (VBA) Syntax

See [Modeler::ImprintingFaces](#).

Remarks

The target and tool faces must:

- belong to different bodies.
- intersect each other.

See Also

[IModeler Interface](#)
[IModeler Members](#)
[IModeler::ImprintingFaces Method](#)
[IModeler::ImprintingFacesCount2 Method](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

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IReplaceSurfaces Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : IReplaceSurfaces Method (IModeler)

Obsolete. Superseded by [IModeler::IReplaceSurfaces2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IReplaceSurfaces( _
    ByVal NFaces As System.Integer, _
```

```

ByRef FaceArray As Face, _
ByRef NewSurfArray As Surface, _
ByRef SenseArray As System.Integer, _
ByVal Tolerance As System.Double _
) As System.Boolean

```

Visual Basic (Usage)



```

Dim instance As IModeler
Dim NFaces As System.Integer
Dim FaceArray As Face
Dim NewSurfArray As Surface
Dim SenseArray As System.Integer
Dim Tolerance As System.Double
Dim value As System.Boolean

value = instance.IReplaceSurfaces(NFaces, FaceArray, NewSurfArray, SenseArray, Tolerance)

```

C#

```

System.bool IReplaceSurfaces(
    System.int NFaces,
    ref Face FaceArray,
    ref Surface NewSurfArray,
    ref System.int SenseArray,
    System.double Tolerance
)

```

C++/CLI

```

System.bool IReplaceSurfaces(
    & System.int NFaces,
    & Face^% FaceArray,
    & Surface^% NewSurfArray,
    & System.int% SenseArray,
    & System.double Tolerance
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NFaces

FaceArray

NewSurfArray

SenseArray

Tolerance

Visual Basic for Applications (VBA) Syntax

See [Modeler::IReplaceSurfaces](#).

See Also

[IModeler Interface](#)
[IModeler Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IReplaceSurfaces2 Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : IReplaceSurfaces2 Method (IModeler)

Replaces the surfaces of the given faces.

.NET Syntax

Visual Basic (Declaration)

```
Function IReplaceSurfaces2( _
    ByVal NFaces As System.Integer, _
    ByRef FaceArray As Face2, _
    ByRef NewSurfArray As Surface, _
    ByRef SenseArray As System.Integer, _
    ByVal Tolerance As System.Double _
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModeler
Dim NFaces As System.Integer
Dim FaceArray As Face2
Dim NewSurfArray As Surface
Dim SenseArray As System.Integer
Dim Tolerance As System.Double
Dim value As System.Boolean

value = instance.IReplaceSurfaces2(NFaces, FaceArray, NewSurfArray, SenseArray, Tolerance)
```

C#

```
System.bool IReplaceSurfaces2(
    System.int NFaces,
    ref Face2 FaceArray,
    ref Surface NewSurfArray,
    ref System.int SenseArray,
    System.double Tolerance
)
```

C++/CLI

```
System.bool IReplaceSurfaces2(
    & System.int NFaces,
    & Face2^% FaceArray,
    & Surface^% NewSurfArray,
    & System.int% SenseArray,
    & System.double Tolerance
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NFaces

Number of faces to have surfaces replaced

FaceArray

Array of [faces](#) to have the surfaces replaced

NewSurfArray

Array of [surfaces](#) to be replaced in these faces

SenseArray

Array of the senses of each surface in NewSurfArray

Tolerance

Tolerance

Return Value

True if operation is successful, false if not

Visual Basic for Applications (VBA) Syntax

See [Modeler.IReplaceSurfaces2](#).

See Also

[IModeler Interface](#)

[IModeler Members](#)

[IModeler.ReplaceSurfaces Method](#)

Availability

IRestore Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : IRestore Method (IModeler)

Obsolete. Superseded by [IModeler::IRestore2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IRestore( _  
    ByVal StreamIn As System.Object _  
) As Body
```

Visual Basic (Usage)



```
Dim instance As IModeler  
Dim StreamIn As System.Object  
Dim value As Body
```

```
value = instance.IRestore(StreamIn)
```

C#

```
Body IRestore(  
    System.object StreamIn  
)
```

C++/CLI

```
Body^ IRestore(  
& System.Object^ StreamIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

StreamIn

Visual Basic for Applications (VBA) Syntax

See [Modeler::IRestore](#).

See Also

[IModeler Interface](#)

[IModeler Members](#)

IRestore2 Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : IRestore2 Method (IModeler)

Restores a temporary body object from the specified stream.

.NET Syntax

Visual Basic (Declaration)

```
Function IRestore2( _  
    ByVal StreamIn As System.Object _  
) As Body2
```

Visual Basic (Usage)



```
Dim instance As IModeler
Dim StreamIn As System.Object
Dim value As Body2

value = instance.IRestore2(StreamIn)
```

C#

```
Body2 IRestore2(
    System.object StreamIn
)
```

C++/CLI

```
Body2^ IRestore2(
    & System.Object^ StreamIn
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

StreamIn
IStream interface for storage inside the SOLIDWORKS document

Return Value

Temporary [body](#)

Visual Basic for Applications (VBA) Syntax

See [Modeler.IRestore2](#).

Remarks

You can use a temporary body object for display purposes. You can generate temporary bodies using such methods as [IModeler::CreateBodyFromFaces2](#) or [IModeler::ICreateBodyFromFaces3](#). Temporary bodies are not saved with the document unless you explicitly save them using [IBody2::Save](#) or [IBody2::ISave](#).

You can obtain the StreamIn object by calling [IModelDoc2::IGet3rdPartyStorage](#) and by specifying the stream name that was used during the body save operation.

See Also

[IModeler Interface](#)
[IModeler Members](#)
[IModeler::Restore Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ISplitFaceOnParam Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : ISplitFaceOnParam Method (IModeler)

Obsolete. Superseded by [IModeler::ISplitFaceOnParam2](#)

.NET Syntax

Visual Basic (Declaration)

```
Function ISplitFaceOnParam() As Face
```

Visual Basic (Usage)



```
Dim instance As IModeler
Dim value As Face

value = instance.ISplitFaceOnParam()
```

C#

```
Face ISplitFaceOnParam()
```

C++/CLI

```
Face^ ISplitFaceOnParam();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [Modeler::ISplitFaceOnParam](#).

See Also

[IModeler Interface](#)
[IModeler Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

ISplitFaceOnParam2 Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : ISplitFaceOnParam2 Method (IModeler)

Splits and retrieves the faces on the U or V parameter.

.NET Syntax

Visual Basic (Declaration)

```
Function ISplitFaceOnParam2() As Face2
```

Visual Basic (Usage)



```
Dim instance As IModeler
Dim value As Face2

value = instance.ISplitFaceOnParam2()
```

C#

```
Face2 ISplitFaceOnParam2()
```

C++/CLI

```
Face2^ ISplitFaceOnParam2();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of new [faces](#)

Visual Basic for Applications (VBA) Syntax

See [Modeler::ISplitFaceOnParam2](#).

Remarks

The split is defined by calling [IModeler::ISplitFaceOnParamCount2](#). Then call this method to retrieve the list of new faces allocated.

See Also

[IModeler Interface](#)
[IModeler Members](#)
[IModeler.SplitFaceOnParam Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

ISplitFaceOnParamCount Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : ISplitFaceOnParamCount Method (IModeler)

Obsolete. Superseded by [IModeler::ISplitFaceOnParamCount2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function ISplitFaceOnParamCount( _  
    ByVal Facedisp As Face, _  
    ByVal UVFlag As System.Integer, _  
    ByVal Parameter As System.Double, _  
    ByRef Status As System.Boolean _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IModeler  
Dim Facedisp As Face  
Dim UVFlag As System.Integer  
Dim Parameter As System.Double  
Dim Status As System.Boolean  
Dim value As System.Integer  
  
value = instance.ISplitFaceOnParamCount(Facedisp, UVFlag, Parameter, Status)
```

C#

```
System.int ISplitFaceOnParamCount(  
    Face Facedisp,  
    System.int UVFlag,  
    System.double Parameter,  
    out System.bool Status  
)
```

C++/CLI

```
System.int ISplitFaceOnParamCount(  
& Face^ Facedisp,  
& System.int UVFlag,  
& System.double Parameter,  
& [Out] System.bool Status  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Facedisp

UVFlag

Parameter

Status

Visual Basic for Applications (VBA) Syntax

See [Modeler::ISplitFaceOnParamCount](#).

See Also

[IModeler Interface](#)
[IModeler Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

ISplitFaceOnParamCount2 Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : ISplitFaceOnParamCount2 Method (IModeler)

Sets up and counts the number of new faces split on the U or V parameter.

.NET Syntax

Visual Basic (Declaration)

```
Function ISplitFaceOnParamCount2( _  
    ByVal Facedisp As Face2, _  
    ByVal UVFlag As System.Integer, _  
    ByVal Parameter As System.Double, _  
    ByRef Status As System.Boolean _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IModeler  
Dim Facedisp As Face2  
Dim UVFlag As System.Integer  
Dim Parameter As System.Double  
Dim Status As System.Boolean  
Dim value As System.Integer  
  
value = instance.ISplitFaceOnParamCount2(Facedisp, UVFlag, Parameter, Status)
```

C#

```
System.int ISplitFaceOnParamCount2(  
    Face2 Facedisp,  
    System.int UVFlag,  
    System.double Parameter,  
    out System.bool Status  
)
```

C++/CLI

```
System.int ISplitFaceOnParamCount2(  
& Face2^ Facedisp,  
& System.int UVFlag,  
& System.double Parameter,  
& [Out] System.bool Status  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Facedisp

[Face](#) to split

UVFlag

Parametric axis; either swSplitFaceOnParamU or swSplitFaceOnParamV

Parameter

Position along the parametric axis at which the split will be performed

Status

True if the operation was successful, false if not

Return Value

Number of new faces

Visual Basic for Applications (VBA) Syntax

See [Modeler::ISplitFaceOnParamCount2](#).

Remarks

The split is defined by calling this method. Then, you can retrieve the list of new faces by using [IModeler::ISplitFaceOnParam2](#).

See Also

[IModeler Interface](#)
[IModeler Members](#)
[IModeler::SplitFaceOnParam Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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MakeManifoldBodies Method (IModeler)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : MakeManifoldBodies Method (IModeler)

Creates manifold bodies from the specified non-manifold body.

.NET Syntax

Visual Basic (Declaration)

```
Function MakeManifoldBodies( _  
    ByVal NonManifoldIn As Body2 _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModeler  
Dim NonManifoldIn As Body2  
Dim value As System.Object  
  
value = instance.MakeManifoldBodies(NonManifoldIn)
```

C#

```
System.object MakeManifoldBodies(  
    Body2 NonManifoldIn  
)
```

C++/CLI

```
System.Object^ MakeManifoldBodies(  
& Body2^ NonManifoldIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NonManifoldIn
Non-manifold [bodies](#)

Return Value

[Manifold bodies](#)

Visual Basic for Applications (VBA) Syntax

See [Modeler::MakeManifoldBodies](#).

Example

[Create and Convert Non-manifold Bodies \(C#\)](#)
[Create and Convert Non-manifold Bodies \(VB.NET\)](#)

See Also

[IModeler Interface](#)
[IModeler Members](#)
[IModeler::IMakeManifoldBodies Method](#)
[IModeler::GeneralTopology Property](#)

Availability

SOLIDWORKS 2006 FCS, Revision Number 14.0